

REV 2026

Practicum

Semester II

Credits 4.5

Contact Hours 90

# Chemistry for Technical Practices

Course Code: 2003B · Polytechnic Diploma (Polymer Technology, Printing Technology, Textile Technology, Chemical Engineering)

|                        |                                                                                   |
|------------------------|-----------------------------------------------------------------------------------|
| <b>Programme(s)</b>    | Polymer Technology, Printing Technology, Textile Technology, Chemical Engineering |
| <b>Subject Code</b>    | 2003B                                                                             |
| <b>Semester</b>        | II                                                                                |
| <b>Course Type</b>     | Practicum                                                                         |
| <b>Teaching Scheme</b> | L:T:P:R = 3:0:3:0                                                                 |
| <b>Contact Hours</b>   | 90                                                                                |
| <b>Credits</b>         | 4.5                                                                               |
| <b>CIE Marks</b>       | 40                                                                                |
| <b>ESE Marks</b>       | 60                                                                                |



## Official Source Declaration

This handbook is based on the Kerala Polytechnic **Revision 2026** syllabus for **Course 2003B - Chemistry for Technical Practices**.

**Verified inconsistencies in source PDF:** The syllabus table for Module IV in the "Detailed Syllabus" section lists "Experiments based on estimation of metals in alloys and preparation of polymers" under both Module IV and Module I subtopics. The handbook treats this as a typographical duplication and places the experimental content under the correct module (Module IV) as per the overall subject flow. Additionally, the CO-PO mapping shows partial correlations; all listed values are reproduced as given.

*All content is elaborated for educational purposes; the syllabus is the authoritative source.*



# Course Dashboard

**4**

Modules

**90**

Total Contact Hours

**90**

Self-Learning Hours

**4.5**

Credits

## Teaching and Assessment Scheme

| Component             | Details                                                                                 |
|-----------------------|-----------------------------------------------------------------------------------------|
| Teaching Scheme       | L:T:P:R = 3:0:3:0                                                                       |
| Contact Hours         | 90                                                                                      |
| Self-Learning Hours   | 90 (as per weekly activities)                                                           |
| Credits               | 4.5                                                                                     |
| CIE Marks             | 40                                                                                      |
| ESE Marks             | 60                                                                                      |
| Assessment Components | Attendance, CA1-CA5 (Self-learning, practical tests, series exams), ESE (Written + Lab) |



# How to Use This Handbook

## 1. Understand

Read each module's concepts and explanations. Use the learning goals and key facts to focus.

## 2. Visualise

Study the labelled diagrams and tables. Use the Visual Library for quick reference.

## 3. Apply

Work through the examples, calculators, and numerical problems. Test yourself with module-wise Q&A.

## 4. Revise

Use the Quick Revision cards, formula bank, glossary, and model paper before exams.



**Tip:** Use the **Print / Download Notes** button to generate a printable PDF for offline study.



# Course Objectives & Outcomes

## Objectives

1. To impart knowledge about fundamental concepts in chemistry including atoms, molecules, solutions, thermodynamics and relate them to engineering applications.
2. To apply key concepts of analytical chemistry and water treatment methods for solving real-world domestic and industrial problems.
3. To impart basic knowledge about organic functional groups, dyes and pigments.

4. To familiarize students with various engineering materials for selecting appropriate materials in industrial and domestic applications and understand green chemistry principles.

## Course Outcomes (CO)

### Course Outcomes and Bloom's Taxonomy Level

| CO  | Course Outcome                                                                                                                                                    | Bloom's Level |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| CO1 | Apply the basic concepts of atoms, molecules, solutions and thermodynamic processes in scientific and engineering contexts.                                       | Apply         |
| CO2 | Apply the fundamentals of analytical Chemistry, appropriate water treatment methods and also the basic concepts of acids and bases to solve engineering problems. | Apply         |
| CO3 | Examine the structure, classification, and applications of dyes, pigments and organic functional groups through theoretical and experimental approaches.          | Apply         |
| CO4 | Explore various engineering materials for suitable industrial and domestic applications and explain green Chemistry principles promoting sustainable practices.   | Apply         |

### CO-PO Mapping (as per syllabus)

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 2   | -   | -   | -   | -   | -   | -   | -   | -    | -    |
| CO2 | 3   | 3   | 2   | 3   | 2   | 2   | 2   | -   | -   | -    | -    |
| CO3 | 2   | 2   | 2   | 2   | 2   | -   | -   | -   | -   | -    | -    |
| CO4 | 3   | 2   | 3   | 2   | 2   | 2   | 3   | -   | -   | -    | -    |

Legends: 1-Low; 2-Medium; 3-High; "-" No Correlation.



## Curriculum Coverage Map



**Module → Topics → Hours → CO**

| Module | Topics                                                                                                                                 | Hours | CO  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| I      | Atoms, Molecules, Solutions, Thermodynamics, Preparation of standard solutions, Endothermic/Exothermic reactions                       | ~22   | CO1 |
| II     | Volumetric analysis, Hardness of water, Water treatment, pH/pOH, Quantitative/Qualitative analysis                                     | ~24   | CO2 |
| III    | Organic functional groups, Dyes (classification, dye-fibre interactions), Pigments, Esterification, tests for unsaturation/aromaticity | ~16   | CO3 |
| IV     | Alloys, Polymers, Nanomaterials, MOFs, Biomaterials, Biofuel, Green Chemistry, Estimation of metals in alloys, preparation of polymers | ~28   | CO4 |



## Module Roadmap



### Module I

Fundamentals: atoms, molecules, solutions, thermodynamics. Builds foundation for all chemical engineering applications.



### Module II

Analytical chemistry & water treatment. Directly applicable to domestic/industrial problem-solving.



### Module III

Organic chemistry, dyes, pigments. Critical for textile, printing, and polymer industries.



### Module IV

Engineering materials (alloys, polymers, nanomaterials, biomaterials) and green chemistry. Integrates sustainability.

## MODULE I

# Atoms, Molecules, Solutions & Thermodynamics

This module covers the basic building blocks of matter, concentration expressions, and the laws governing energy changes — essential for all engineering chemistry.

### Learning Goals

- Define atom, molecule, solute, solvent.
- Calculate fundamental particles from atomic number/mass number.
- Solve molarity and normality problems.
- Explain thermodynamic systems, processes, and state functions.

### Key Facts

- $Z = \text{protons} = \text{electrons}$  (neutral atom).
- Molarity (M) = moles solute / L solution.
- Normality (N) = gram-equivalents / L solution.
- Enthalpy (H) =  $U + PV$ ; Entropy (S) = disorder.



## Atoms and Molecules

**Atom:** The smallest particle of an element that retains its chemical properties. Example: Hydrogen atom (H), Oxygen atom (O).

**Molecule:** A group of two or more atoms held together by chemical bonds. Example:  $\text{H}_2\text{O}$  (water),  $\text{O}_2$  (oxygen gas).

**Fundamental particles:** Electron ( $e^-$ , charge -1, mass  $\sim 9.1 \times 10^{-31}$  kg), Proton ( $p^+$ , charge +1, mass  $\sim 1.67 \times 10^{-27}$  kg), Neutron ( $n^0$ , charge 0, mass  $\sim 1.67 \times 10^{-27}$  kg).

**Atomic number (Z):** Number of protons in the nucleus. **Mass number (A):** Total number of protons + neutrons.

**Example:** For Calcium (Ca,  $Z=20$ ,  $A=40$ ): protons=20, electrons=20, neutrons= $40-20=20$ .

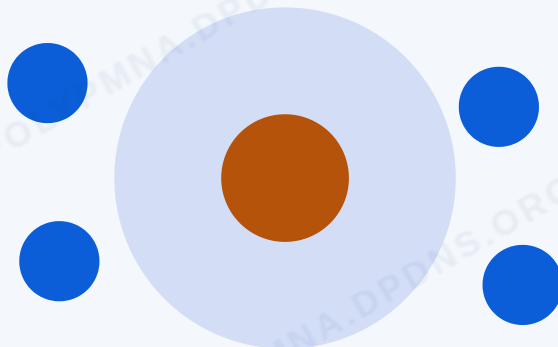


Fig: Schematic atom showing nucleus (protons+neutrons) and orbiting electrons.

**Malayalam support:** ആറ്റം (atom) മൂലകത്തിന്റെ ഏറ്റവും ചെറിയ കണികയാണ്. തന്മാത്ര (molecule) രണ്ടോ അതിലധികമോ ആറ്റങ്ങൾ ചേർന്നതാണ്. ഇലക്ട്രോൺ, പ്രോട്ടോൺ, ന്യൂട്രോൺ എന്നിവയാണ് അടിസ്ഥാന കണങ്ങൾ.

## Solutions and Concentration Expressions

**Solute:** Substance that dissolves (e.g., salt). **Solvent:** Substance that dissolves the solute (e.g., water). **Standard solution:** A solution of known concentration.

**Molarity (M)** = moles of solute / volume of solution (L)

**Normality (N)** = gram-equivalents of solute / volume of solution (L)

**ppm** = (mass of solute / mass of solution)  $\times 10^6$

**Worked example (Molarity):** 58.5 g of NaCl (M.W. = 58.5 g/mol) dissolved in water to make 1 L solution.  $M = 58.5/58.5 / 1 = 1 \text{ M}$ .



### Molarity Calculator

Calculate molarity from mass, molecular weight, and volume.

Mass (g):

Molecular weight (g/mol):

Volume (L):

Molarity = 1.00 M

## Thermodynamics: Systems, Processes, State Functions

**System:** The part of the universe under study. **Surroundings:** Everything else.

**Types of systems:** Open (mass & energy exchange), Closed (energy only), Isolated (neither).

**Processes:** Isothermal (constant T), Isobaric (constant P), Isochoric (constant V), Adiabatic (no heat exchange).

**Reversible vs. Irreversible:** Reversible proceeds infinitesimally slowly; irreversible is spontaneous.

**State functions:** Internal energy (U), Enthalpy ( $H = U + PV$ ), Entropy (S).

### Comparison of Thermodynamic Processes

| Process    | Constant         | Example                                           |
|------------|------------------|---------------------------------------------------|
| Isothermal | Temperature      | Slow expansion of gas in contact with a heat bath |
| Isobaric   | Pressure         | Heating gas in a cylinder with a movable piston   |
| Isochoric  | Volume           | Heating gas in a rigid container                  |
| Adiabatic  | No heat exchange | Rapid compression of gas in a insulated cylinder  |

**Tip:** Endothermic reactions absorb heat ( $\Delta H > 0$ ); exothermic release heat ( $\Delta H < 0$ ).

### Practical: Preparation of Standard Solutions & Endothermic/Exothermic Reactions

**Preparation of 0.1 N oxalic acid solution:** Weigh 6.3 g of oxalic acid ( $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ ) and dissolve in distilled water to make 1 L.

**Endothermic reaction:** Citric acid + baking soda (absorbs heat, feels cold).

**Exothermic reaction:** Neutralisation  $\text{HCl} + \text{NaOH}$  (releases heat, feels warm).

**Safety:** Wear gloves and goggles when handling acids and bases. Use a fume hood if required.

## MODULE II

# Analytical Chemistry & Water Treatment

 Contact Hours: ~24

 CO2

 Bloom: Apply

Understand volumetric analysis, water hardness, treatment methods, pH concepts, and their application in solving engineering problems.

## Volumetric Analysis

**Titration:** A technique where a solution of known concentration (titrant) is added to a solution of unknown concentration (analyte) until the reaction is complete (end point).

**Normality equation:**  $N_1V_1 = N_2V_2$

**Example:** 20 mL of 0.1 N HCl neutralises 40 mL of NaOH. Find normality of NaOH.  
 $N_1V_1 = N_2V_2 \rightarrow 0.1 \times 20 = N_2 \times 40 \rightarrow N_2 = 0.05 \text{ N}$

### Choice of Indicators

| Titration type            | Indicator                       | pH range         |
|---------------------------|---------------------------------|------------------|
| Strong acid – Strong base | Phenolphthalein / Methyl orange | 8.2–10 / 3.1–4.4 |
| Strong acid – Weak base   | Methyl orange                   | 3.1–4.4          |
| Weak acid – Strong base   | Phenolphthalein                 | 8.2–10           |

## Hardness of Water & Treatment Methods

**Hard water:** Contains dissolved salts of  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$  (bicarbonates, chlorides, sulphates).

**Temporary hardness:**  $\text{Ca}(\text{HCO}_3)_2$ ,  $\text{Mg}(\text{HCO}_3)_2$  (removed by boiling). **Permanent hardness:**  $\text{CaCl}_2$ ,  $\text{MgCl}_2$ ,  $\text{CaSO}_4$ ,  $\text{MgSO}_4$  (removed by chemical treatment).

**Municipal water treatment steps:** Screening → Sedimentation → Coagulation → Filtration → Sterilisation.

**Sterilisation methods:** Chlorination, UV radiation, Ozonation.

**Advanced treatment:** Reverse osmosis, Nano-filtration.

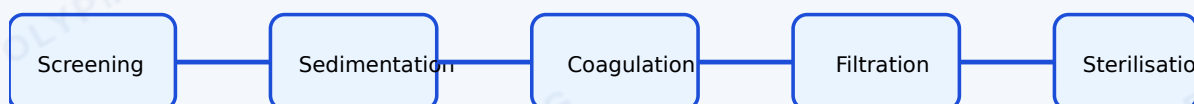


Fig: Municipal water treatment flow chart.

## Acids, Bases, pH and pOH

**Arrhenius concept:** Acid produces  $\text{H}^+$  in water; base produces  $\text{OH}^-$  in water.

$\text{pH} = -\log_{10}[\text{H}^+]$ ,  $\text{pOH} = -\log_{10}[\text{OH}^-]$ ,  $\text{pH} + \text{pOH} = 14$  (at  $25^\circ\text{C}$ ).

**Example:**  $[\text{H}^+] = 1 \times 10^{-4} \text{ M} \rightarrow \text{pH} = 4$  (acidic).  $[\text{OH}^-] = 1 \times 10^{-3} \text{ M} \rightarrow \text{pOH} = 3 \rightarrow \text{pH} = 11$  (basic).

### pH Calculator

$[\text{H}^+]$  (M):

$\text{pH} = 4.00$

**Practical: Volumetric Analysis & pH Determination**

- Standardisation of HCl using  $\text{Na}_2\text{CO}_3$ .
- Estimation of NaOH using standard HCl.
- Estimation of KOH using standard oxalic acid.
- Standardisation of EDTA using  $\text{ZnSO}_4$ .
- Volumetric estimation of total hardness of water using EDTA.
- Determination of pH using pH meter, universal indicator, pH paper.

**MODULE III****Organic Functional Groups, Dyes & Pigments**

Contact Hours: ~16



CO3



Bloom: Apply

Explore the structure, classification, and applications of organic functional groups, dyes, and pigments — essential for textile and printing technologies.

**Organic Functional Groups**

**Functional group:** A specific group of atoms within a molecule that is responsible for its characteristic chemical reactions.

- **Alcohol:** -OH (e.g., ethanol)
- **Acid:** -COOH (e.g., acetic acid)
- **Ester:** -COOR (e.g., ethyl acetate)
- **Hydrocarbons:** Aliphatic (linear/branched) and Aromatic (benzene ring); Saturated (single bonds) and Unsaturated (double/triple bonds).

**Examples of Functional Groups**

| Group           | Structure | Example                                                  |
|-----------------|-----------|----------------------------------------------------------|
| Alcohol         | -OH       | Ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ )            |
| Carboxylic acid | -COOH     | Acetic acid ( $\text{CH}_3\text{COOH}$ )                 |
| Ester           | -COO-     | Ethyl acetate ( $\text{CH}_3\text{COOCH}_2\text{CH}_3$ ) |



## **Dyes: Classification, Chromophore, Auxochrome**

**Dye:** A coloured substance that imparts colour to a material.

**Chromophore:** The part of the molecule responsible for colour (e.g.,  $-N=N-$ ,  $-NO_2$ ).

**Auxochrome:** A group that intensifies colour and helps in bonding (e.g.,  $-OH$ ,  $-NH_2$ ).

**Classification by source:** Natural (e.g., indigo) and Synthetic (e.g., azo dyes).

**Classification by application:** Direct, Acid, Basic, Disperse, Mordant, Vat dyes.

**Classification by structure:** Azo, Anthraquinone, Nitro, Indigo dyes.

**Dye-fibre interactions:** Physical adsorption, Hydrogen bonding, Ionic, Covalent interactions.

**Tip:** Direct dyes are applied directly to cotton; vat dyes are insoluble and require reduction.

## **Pigments**

**Pigment:** A substance that imparts colour by reflecting light; usually insoluble and dispersed in a medium.

**Types:** Inorganic (e.g., Titanium dioxide, Iron oxide) and Organic (e.g., Phthalocyanine).

**Uses:** Paints, inks, plastics, cosmetics, textiles.

## **Practical: Preparation of Dyes, Esterification, Identification Tests**

- **Preparation of azo dye:** Diazotisation and coupling reaction.
- **Esterification reaction:** Alcohol + carboxylic acid  $\rightarrow$  ester + water (with acid catalyst).
- **Identification tests:** Alcohol (ceric ammonium nitrate test), Acid (litmus test), Ester (hydroxamic acid test).
- **Test for unsaturation:** Baeyer's test ( $KMnO_4$  decolourisation).
- **Test for aromaticity:** Baeyer's test (resistance to  $KMnO_4$ ) and ignition test (sooty flame).

MODULE IV

# Engineering Materials & Green Chemistry



Contact Hours: ~28



CO4



Bloom: Apply

Study the properties, applications, and sustainability aspects of alloys, polymers, nanomaterials, biomaterials, and green chemistry.



## Alloys

**Alloy:** A mixture of two or more metals (or a metal and a non-metal).

**Purposes of alloying:** To increase strength, corrosion resistance, hardness, and to improve appearance.

### Common Alloys and Applications

| Alloy           | Composition  | Applications                        |
|-----------------|--------------|-------------------------------------|
| Stainless steel | Fe + Cr + Ni | Kitchenware, surgical instruments   |
| Brass           | Cu + Zn      | Decorative items, plumbing fittings |
| Bronze          | Cu + Sn      | Statues, bearings                   |
| Solder          | Pb + Sn      | Joining electrical wires            |

 **Polymers**

**Monomer:** Small repeating unit. **Polymer:** Large molecule made of many monomers.

**Polymerisation:** The process of forming polymers.

**Homo vs Copolymer:** Homopolymer: one type of monomer (e.g., PE). Copolymer: two or more types (e.g., Buna-S).

**Thermoplastics:** Soften on heating (e.g., PE, PVC, Nylon-66). **Thermosetting plastics:** Set permanently on heating (e.g., Bakelite).

**Common Polymers and Applications**

| Polymer  | Monomer                             | Applications                 |
|----------|-------------------------------------|------------------------------|
| PE       | Ethylene                            | Plastic bags, containers     |
| PVC      | Vinyl chloride                      | Pipes, cables                |
| Nylon-66 | Hexamethylene diamine + Adipic acid | Fibres, ropes                |
| Bakelite | Phenol + Formaldehyde               | Electrical switches, handles |
| PET      | Ethylene glycol + Terephthalic acid | Bottles, fibres              |
| PU       | Diisocyanate + Polyol               | Foams, coatings              |
| PMMA     | Methyl methacrylate                 | Transparent sheets, lenses   |

**Natural rubber:** Monomer is isoprene. **Vulcanisation:** Cross-linking with sulphur to improve strength, elasticity, and heat resistance.

**Synthetic rubber:** Buna-S (Styrene + Butadiene), Buna-N (Acrylonitrile + Butadiene).



## Nanomaterials, MOFs, Biomaterials, Biofuel, Green Chemistry

**Nanomaterials:** Materials with at least one dimension in the 1–100 nm range. Examples: Carbon nanotubes (CNTs), Fullerenes (C<sub>60</sub>), Graphene (single layer of carbon). Applications: Electronics, medicine, composites.

**Metal-Organic Frameworks (MOFs):** Porous materials made of metal ions and organic linkers. Applications: Gas storage, catalysis, drug delivery.

**Biomaterials:** Materials used in biomedical applications. Types: Metallic (e.g., Ti implants), Ceramic (e.g., bone grafts), Polymeric (e.g., sutures), Composite (e.g., dental fillings).

**Biofuel:** Fuel produced from biological sources (e.g., ethanol from corn, biodiesel from vegetable oils). Applications: Transportation, power generation.

**Green Chemistry:** Design of chemical products and processes that reduce or eliminate hazardous substances. **Principles:** Prevention, Atom economy, Less hazardous synthesis, Safer solvents, Energy efficiency, Renewable feedstocks, Catalysis, etc.



## Practical: Estimation of Metals in Alloys & Polymer Preparation

- Estimation of copper in brass.
- Estimation of zinc in brass.
- Estimation of iron in iron ore.
- Preparation of Urea-formaldehyde resin.
- Preparation of Phenol-formaldehyde resin.



## Formula Bank

### Molarity

$$M = \text{moles solute} / \text{L solution}$$

### Normality

$$N = \text{gram-equivalents} / \text{L solution}$$

**ppm**

$$(\text{mass solute} / \text{mass solution}) \times 10^6$$

**Normality equation**

$$N_1V_1 = N_2V_2$$

**pH**

$$\text{pH} = -\log_{10}[\text{H}^+]$$

**pOH**

$$\text{pOH} = -\log_{10}[\text{OH}^-]$$

**pH + pOH**

$$= 14 \text{ (at } 25^\circ\text{C)}$$

**Enthalpy**

$$H = U + PV$$

**Entropy**

S = disorder measure



## Visual Library



### Atom Model

[See Module I](#) - Nucleus with protons/neutrons and orbiting electrons.



### Water Treatment Flow

[See Module II](#) - Screening → Sedimentation → Coagulation → Filtration → Sterilisation.



### Dye-Fibre Interactions

[See Module III](#) – Physical adsorption, H-bonding, ionic, covalent.

### Polymer Types

[See Module IV](#) – Thermoplastics vs Thermosetting plastics.



## Self-Learning Activities

### Module I

- Assignment: atomic number/mass number of 20 elements.
- Model of atom using chart/paper.
- Report on concentration terms.
- Observe endothermic/exothermic processes at home.

### Module II

- Problems on normality equation.
- Natural indicators from household materials.
- Collect water samples and identify pH.
- Report on water purification methods.

### Module III

- Report on natural and synthetic dyes in textile printing.
- Video on dyeing clothes.
- Short note on esters in daily life.
- Report on pigments used in textile/printing.

### Module IV

- Assignment on articles made of alloys.
- Report on bioplastics.
- Short video on nanomaterials.

- Report on biomaterials and MOFs.
- Assignment on biofuels.



## Assessment Methodology

### Assessment Components

| Mode       | Description                                          | Marks   | Converted to |
|------------|------------------------------------------------------|---------|--------------|
| Attendance | -                                                    | -       | 5            |
| CA1        | Average of Self-learning activities from each module | 50      | 10           |
| CA2        | Practical test (first half)                          | 40      | 10           |
| CA3        | Practical test (second half)                         | 40      | 10           |
| CA4        | Series Exam (2 modules)                              | 50      | 10           |
| CA5        | Series Exam (2 modules)                              | 50      | 10           |
| ESE        | Written Exam (theory) + Lab Examination              | 60 + 40 | 60           |

### Series Exam Pattern (2 hours)

Part A: 2 questions  $\times$  1 mark = 2

Part B: 3 questions  $\times$  3 marks = 9

Part C: 1 set (with choice)  $\times$  7 marks = 7

Total: 18 marks (per module, 2 modules combined)

### ESE Pattern (2.5 hours)

Part A: 2 questions/module  $\times$  1 mark = 8

Part B: 2 out of 3/module  $\times$  3 marks = 24

Part C: 1 set with choice/module  $\times$  7 marks = 28

Total: 60 marks

**Note:** Minimum of 10 experiments from all modules must be performed.

## ? Module-wise Questions & Answers

### Module I

#### 1. Define atom and molecule with examples.

Atom is the smallest particle of an element that retains its properties (e.g., H). Molecule is a group of atoms bonded together (e.g., H<sub>2</sub>O).

#### 2. Calculate the number of protons, neutrons and electrons in an atom with Z=17, A=35.

Protons=17, electrons=17, neutrons=35-17=18.

#### 3. What is molarity? Calculate the molarity of a solution containing 40 g NaOH (M.W. 40) in 2 L water.

$M = \text{moles/L} = (40/40)/2 = 0.5 \text{ M}.$

#### 4. Distinguish between isothermal and adiabatic processes.

Isothermal: constant temperature; Adiabatic: no heat exchange.

### Module II

#### 1. State the principle of volumetric analysis.

At the equivalence point, equivalents of acid = equivalents of base ( $N_1V_1=N_2V_2$ ).



**2. Differentiate between temporary and permanent hardness of water.**

Temporary: caused by bicarbonates, removed by boiling. Permanent: caused by chlorides/sulphates, removed by chemical treatment.

**3. Draw a flow chart for municipal water treatment.**

Screening → Sedimentation → Coagulation → Filtration → Sterilisation.

**4. Calculate pH of a solution with  $[H^+] = 1 \times 10^{-6} \text{ M}$ .**

$\text{pH} = -\log(10^{-6}) = 6.$

**Module III****1. Define chromophore and auxochrome with examples.**

Chromophore: colour-producing group (e.g.,  $-N=N-$ ). Auxochrome: intensifies colour (e.g.,  $-OH$ ).

**2. Classify dyes based on method of application.**

Direct, Acid, Basic, Disperse, Mordant, Vat.

**3. What are pigments? Give two examples.**

Insoluble colouring substances; e.g.,  $\text{TiO}_2$  (inorganic), Phthalocyanine (organic).

**4. Describe the test for unsaturation.**

Baeyer's test: addition of  $\text{KMnO}_4$  (purple) → decolourisation indicates unsaturation.

**Module IV****1. Define alloy. List the purposes of alloying.**

Alloy is a mixture of metals. Purposes: increase strength, corrosion resistance, hardness.

**2. Differentiate between thermoplastics and thermosetting plastics.**

Thermoplastics soften on heating (e.g., PE); thermosetting set permanently (e.g., Bakelite).

**3. What is vulcanisation? List properties of vulcanised rubber.**

Vulcanisation: cross-linking rubber with sulphur. Properties: stronger, more elastic, heat resistant.

**4. State any three principles of green chemistry.**

Prevention, Atom economy, Safer solvents.



## Practice Model Paper

*Based on the official examination pattern. Not an official SITTTTR model question paper.*

**Time: 2.5 hours | Max Marks: 60**

**Part A (1 mark each - 2 questions per module)**

- 1. Define atomic number.
- 2. What is ppm?
- 3. State the normality equation.
- 4. What is the pH of pure water?
- 5. Define functional group.
- 6. What is a chromophore?
- 7. Define alloy.
- 8. What is vulcanisation?

**Part B (3 marks each - 2 out of 3 per module)**

- 1. Explain molarity and normality with formulas.
- 2. Distinguish between open and closed systems.
- 3. Explain the steps in municipal water treatment.
- 4. What is the choice of indicator in acid-base titrations?
- 5. Classify dyes based on chemical structure.
- 6. Explain the esterification reaction.
- 7. List the applications of common polymers.
- 8. Explain the principles of green chemistry.

**Part C (7 marks each - 1 set with choice per module)**

- 1. a) Calculate the number of particles in an atom with  $Z=20$ ,  $A=40$ . b) Solve a molarity problem.
- 2. a) Explain hardness of water and its removal. b) Calculate pH from  $[H^+]$ .
- 3. a) Discuss the classification of dyes. b) Explain dye-fibre interactions.
- 4. a) Explain polymers and their applications. b) Discuss nanomaterials and green chemistry.

## ⚡ Quick Revision

**Module I**

- $Z = \text{protons} = \text{electrons (neutral)}$
- $M = \text{moles/L}; N = \text{eq/L}$
- $\Delta H = H(\text{products}) - H(\text{reactants})$

**Module II**

- $N_1V_1 = N_2V_2$
- $\text{pH} = -\log[H^+]$
- Water treatment: screening  $\rightarrow$  sedimentation  $\rightarrow$  coagulation  $\rightarrow$  filtration  $\rightarrow$  sterilisation

**Module III**

- Chromophore = colour; Auxochrome = intensifier
- Dye classification: application (direct, acid, etc.) and structure (azo, anthraquinone, etc.)

**Module IV**

- Alloying improves properties
- Thermoplastics vs Thermosetting
- Green Chemistry: 12 principles

**⚠ Common mistakes:** Confusing molarity and normality; forgetting units in pH calculations; misclassifying dyes; not distinguishing between thermoplastics and thermosetting.



## Glossary

| Term            | Meaning                                             |
|-----------------|-----------------------------------------------------|
| Atom            | Smallest particle of an element                     |
| Molecule        | Group of atoms bonded together                      |
| Molarity        | Moles of solute per litre of solution               |
| Normality       | Gram-equivalents of solute per litre of solution    |
| pH              | Negative logarithm of $H^+$ concentration           |
| Alloy           | Mixture of metals                                   |
| Polymer         | Large molecule made of repeating monomers           |
| Chromophore     | Colour-producing group in a dye                     |
| Green Chemistry | Design of environmentally benign chemical processes |



## References

### Textbooks

| Title                                              | Author                    |
|----------------------------------------------------|---------------------------|
| Principles of Physical Chemistry                   | Puri, Sharma & Pathania   |
| Vogel's Textbook of Quantitative Chemical Analysis | Arthur Israel Vogel       |
| Fundamentals of Analytical Chemistry               | Skoog, West, Holler       |
| Nanoscience and Technology                         | Muralidharan & Subramania |
| Color Chemistry                                    | Heinrich Zollinger        |

**Web Resource:** [NPTEL – Chemistry](#)



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